

Leakage-controlled stacking and recursive feature elimination for type 2 diabetes prediction: a rigorous evaluation on the PIMA dataset

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5 Abstract

Background: Machine-learning models for type 2 diabetes mellitus (T2DM) screening often report optimistic performance because imputation or feature selection is fitted before validation, and rarely test differences between models.

10 **Methods:** We built a leakage-controlled pipeline combining median imputation of impossible zero values, eight engineered clinical features, recursive feature elimination with cross-validation (RFECV) and an eight-learner stacking ensemble with a logistic-regression meta-learner. Every step was re-fitted inside each validation split. On the PIMA Indians Diabetes dataset
15 (768 women, 268 with diabetes), we compared 11 learners with and without a common RFECV subset using 5×10 -fold repeated cross-validation with corrected confidence intervals and Holm-adjusted tests. Results were confirmed on a 154-patient hold-out set with bootstrap intervals and DeLong tests. We also assessed calibration, screening thresholds, decision curves, imputation
20 choices and SHAP explanations.

Results: RFECV reduced inputs by 25% without loss of discrimination, consistently retaining glucose, BMI, family history and their interactions. Cross-validated ROC-AUC was 0.843 (95% CI 0.809–0.878) for RFECV + stacking and 0.842 (0.809–0.875) for logistic regression (adjusted $p = 1.00$).
25 On the hold-out set, RFECV + stacking achieved sensitivity 0.815, specificity 0.740, MCC 0.532 and ROC-AUC 0.822. These were the highest observed sensitivity and MCC, but no ROC-AUC difference was significant. Class weighting caused systematic risk overestimation (calibration intercept -0.62 , slope 1.00). Results were unchanged under iterative, k -NN or indicator-based
30 imputation.

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Conclusions: Under rigorous, leakage-free validation, stacking and RFECV give compact, sensitivity-oriented models but no discrimination gain over logistic regression on this small cohort. Calibration and threshold choice deserve as much attention as model choice.

Keywords: Type 2 diabetes mellitus, Stacked generalisation, Recursive feature elimination, Data leakage, Calibration, Decision curve analysis, PIMA Indians Diabetes

1. Introduction

35 An estimated 536.6 million adults aged 20–79 years were living with diabetes in 2021, and this number is projected to reach 783.2 million by 2045 [1]. Type 2 diabetes mellitus (T2DM) accounts for the large majority of cases, and a substantial share of people with diabetes are undiagnosed [2]. Because hyperglycaemia damages the kidneys, retina, nerves and vasculature years
40 before symptoms appear, inexpensive risk estimates from routinely collected measurements could help target confirmatory testing.

Machine-learning (ML) classifiers trained on variables such as plasma glucose, body-mass index (BMI), blood pressure and family history are a common route to such screening tools. Two design decisions dominate this
45 literature. The first is feature selection: wrapper methods such as recursive feature elimination (RFE) [3, 4] prune redundant inputs, and RFE with cross-validation (RFECV) chooses the subset size by validated performance. The second is model choice: stacked generalisation [5] trains a meta-learner on the out-of-fold predictions of diverse base learners and is often reported
50 to outperform any single learner.

On small clinical datasets, however, the reported gains of such pipelines are hard to interpret. Imputation, scaling, oversampling or feature selection are frequently fitted on the complete dataset before cross-validation or splitting. This leaks information from the evaluation data into model development and inflates performance [6–8]. Moreover, differences between
55 models are usually reported without confidence intervals or paired tests, calibration is rarely assessed [9], and the default 0.5 decision threshold is seldom justified for screening. Systematic comparisons have also found no average performance benefit of flexible ML over logistic regression for clinical prediction [10].
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This study therefore asks a deliberately narrow question: *when every data-dependent step is kept inside the validation loop, does combining RFECV*

with a heterogeneous stacking ensemble improve T2DM prediction on the PIMA cohort over simpler models, and what does it offer for screening? Our
 65 contributions are methodological rather than algorithmic:

1. A leakage-controlled pipeline in which median imputation of physiologically impossible zeros, clinically motivated feature engineering, standardisation, RFECV and an eight-learner stacking ensemble are all re-fitted within the training part of every split.
- 70 2. A common feature-selection protocol in which RFECV is fitted once per outer fold and its subset is shared by every compared learner, together with a learner-specific RFECV sensitivity analysis for logistic regression.
- 75 3. A statistically explicit evaluation: 5×10 -fold repeated cross-validation with corrected confidence intervals and corrected resampled t -tests [11, 12], Holm adjustment for multiplicity [13], bootstrap confidence intervals and DeLong tests [14, 15] on an untouched hold-out set.
- 80 4. A clinically oriented assessment that goes beyond discrimination: calibration, pre-specified sensitivity-targeted thresholds, decision-curve analysis [16], a missing-data sensitivity analysis and SHAP explanations [17].

We report where the framework helps and where it does not. In particular, under strict validation the stacking ensemble does not discriminate better than regularised logistic regression on this cohort.

2. Related work

85 *Machine learning on the PIMA cohort.* The PIMA Indians Diabetes dataset originates from a long-running longitudinal study of the Gila River Indian Community [18] and was introduced as an ML benchmark with the ADAP neural network [19]. It has since been used to evaluate logistic regression, support vector machines, tree ensembles such as random forests [20], XG-Boost [21] and LightGBM [22], and deep networks. Reported accuracies
 90 range from about 0.75 to above 0.98. Recent examples include a Boruta and stacking model reporting 98% accuracy [23] and an RFE-GRU network reporting 90.7% accuracy [24]. By contrast, Reza et al. reported 75.0% (train-test split) and 77.1% (cross-validation) for a stacked model on PIMA [25]. A
 95 spread this wide on a fixed 768-patient dataset is more plausibly explained by differences in validation design than by differences between algorithms.

Stacking and feature selection for diabetes. Wolpert introduced stacked generalisation [5], and Ting and Witten showed that stacking works best when a simple linear meta-learner receives class probabilities [26]. In this journal, 100 Kalagotla et al. combined correlation-based feature selection with a stack of a multilayer perceptron, SVM and logistic regression on PIMA [27]. Idris et al. combined stacking with RFE and isolation-forest outlier removal and reported 79.1% accuracy on PIMA [28]. The combination of RFE-type selection and stacking is therefore not new. What is less common is (i) validating 105 the *whole* chain, including imputation and selection, without leakage; (ii) quantifying uncertainty and testing differences between models; and (iii) reporting calibration and clinical utility alongside discrimination.

Leakage, selection bias and reporting. Ambroise and McLachlan showed that selecting features on the full dataset before cross-validation gives strongly 110 optimistic error estimates [6]. Varma and Simon showed a similar bias when tuning and evaluation share folds [29]. Kaufman et al. formalised data leakage [7], and Kapoor and Narayanan documented leakage across hundreds of ML-based studies in 17 fields [8]. The TRIPOD+AI statement now asks prediction-model studies to report how missing data were handled, how performance was validated, and how calibration and clinical utility were as- 115 sessed [30]. Our evaluation design follows these recommendations.

3. Materials and methods

Fig. 1 summarises the framework. This study is reported following the TRIPOD+AI recommendations where applicable [30].

120 3.1. Data

We used the PIMA Indians Diabetes dataset [19]: 768 women of Pima heritage aged at least 21 years, of whom 268 (34.9%) had diabetes. Each record contains eight predictors: number of pregnancies, 2-h plasma glucose concentration in an oral glucose tolerance test (OGTT; mg/dL), diastolic 125 blood pressure (mm Hg), triceps skin-fold thickness (mm), 2-h serum insulin ($\mu\text{U/mL}$), BMI (kg/m^2), diabetes pedigree function and age (years). The outcome is diabetes onset. The data are publicly available and de-identified.

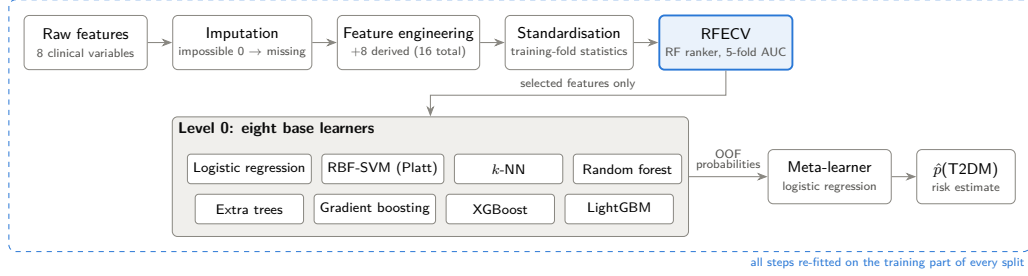


Figure 1: Proposed pipeline. Imputation, feature engineering, standardisation and RFECV are fitted on the training part of each split only. The features retained by RFECV are passed to eight heterogeneous base learners, whose out-of-fold (OOB) probabilities train a logistic-regression meta-learner.

3.2. Missing values

Zero is physiologically impossible for glucose, blood pressure, skin-fold thickness, insulin and BMI, so these zeros were recoded as missing (Table 1). In the primary analysis, missing values were replaced by the median of the training data in each split. Because insulin is missing for almost half of the patients, we also compared three alternatives in a sensitivity analysis (Section 3.9).

Table 1: Physiologically impossible zero values recoded as missing ($n = 768$).

Predictor	Missing (n)	Missing (%)
Insulin	374	48.7
Skin-fold thickness	227	29.6
Diastolic blood pressure	35	4.6
BMI	11	1.4
Glucose	5	0.7

3.3. Feature engineering

Eight derived features encoding known T2DM risk interactions were added after imputation, giving 16 candidates for RFECV (Table 2). The insulin-resistance term follows the HOMA-IR formula [31], but PIMA records 2-h post-load rather than fasting glucose and insulin. We therefore name it

140 HOMA_IR_surrogate and do not interpret it as clinical HOMA-IR. The binary thresholds follow the WHO obesity cut-off [32] and the 2-h OGTT threshold for impaired glucose tolerance [33].

Table 2: Derived features. All continuous inputs are measured after imputation.

Feature	Definition	Rationale
Glucose_BMI	$\text{Glucose} \times \text{BMI} / 100$	Joint glycaemia-adiposity effect
Glucose_Age	$\text{Glucose} \times \text{Age} / 100$	Glycaemic risk rising with age
HOMA_IR_surrogate	$\text{Glucose} \times \text{Insulin} / 405$	Insulin-resistance surrogate (post-load)
Insulin_Glucose_Ratio	$\text{Insulin} / \text{Glucose}$	Insulin response relative to glucose
BMI_Age	$\text{BMI} \times \text{Age} / 100$	Cumulative adiposity exposure
Obese	1 if $\text{BMI} \geq 30$	WHO obesity threshold
Hyperglycemic	1 if $\text{Glucose} \geq 140$	Impaired glucose tolerance threshold
Pedigree_Age	$\text{Pedigree} \times \text{Age}$	Familial risk expressed over time

3.4. Recursive feature elimination with cross-validation

145 All features were standardised with training statistics. RFECV started from the 16 candidates. At each step it fitted a random forest (200 trees, balanced class weights), ranked features by impurity-based importance and removed the least important one. Each subset size was scored by stratified 5-fold cross-validated ROC-AUC within the training data, and the size with the highest mean score was retained (minimum three features).

150 *Common versus learner-specific selection.* In the model comparison, RFECV was fitted once within each outer training fold using the random-forest ranking estimator, and the resulting subset was supplied to every candidate learner. This gives all learners an identical, leakage-free feature-selection protocol at the cost of one RFECV fit per fold. The subset is optimised
155 with the random-forest ranking criterion, not separately for each learner.

It is therefore *not* equivalent to cross-validating a learner-specific RFECV pipeline for every classifier. To gauge the effect of this choice, we also evaluated logistic regression with its own RFECV ranked by absolute standardised coefficients (“LR-ranked RFECV”).

160 3.5. *Stacking ensemble*

The level-0 learners were chosen for diversity of inductive bias (Table 3). The level-1 meta-learner was an L2-regularised logistic regression with balanced class weights. It was trained on the positive-class probabilities produced by each base learner for held-out samples in an internal stratified 5-fold
 165 split [5, 26]. The base learners were then refitted on all training data for prediction. Gaussian naive Bayes and a depth-5 decision tree were evaluated as additional reference learners but were not part of the stack. Hyperparameters were fixed a priori at common defaults (Table 3) and were not tuned, so that no tuning loop could leak into the evaluation.

Table 3: Level-0 learners and their fixed settings.

Learner	Family	Settings
Logistic regression (LR)	Linear	L2, $C = 1$, balanced class weights
SVM	Kernel	RBF, $C = 1$, balanced weights, Platt-calibrated [34]
k -nearest neighbours	Instance-based	$k = 15$, distance-weighted
Random forest [20]	Bagged trees	300 trees, min. leaf 2, balanced weights
Extra trees	Randomised trees	300 trees, min. leaf 2, balanced weights
Gradient boosting	Boosted trees	200 trees, depth 3, learning rate 0.05
XGBoost [21]	Boosted trees	300 trees, depth 3, rate 0.05, subsample 0.8
LightGBM [22]	Boosted trees	300 trees, 15 leaves, rate 0.05, balanced weights

170 3.6. *Validation design*

The data were split once into a stratified 80% development set (614 patients, 214 with diabetes) and a 20% hold-out set (154 patients, 54 with

diabetes), with random seed 42. The hold-out set was not used for any modelling decision.

175 *Repeated cross-validation.* On the development set, all 11 learners were compared with and without the common RFECV subset using stratified 10-fold cross-validation repeated five times (50 folds). Within each fold, imputation, feature engineering, standardisation and RFECV were fitted on the nine training folds only. Because fold estimates from overlapping training
180 sets are correlated, we report the mean of each metric with a 95% confidence interval based on the Nadeau–Bengio corrected variance [11],

$$\widehat{\text{Var}}_{\text{corr}} = \left(\frac{1}{J} + \frac{n_{\text{test}}}{n_{\text{train}}} \right) s^2, \quad (1)$$

where $J = 50$ is the number of fold estimates, s^2 their sample variance, and $n_{\text{test}}/n_{\text{train}} = 1/9$. Each configuration was compared with logistic regression on all features using the corrected resampled t -test on paired fold differences
185 ($J - 1$ degrees of freedom) [11, 12]. p -values were adjusted for the 22 comparisons with the Holm procedure [13].

Hold-out evaluation. The final pipelines were fitted on the full development set and scored once on the hold-out set. ROC-AUC confidence intervals were computed with DeLong’s method, and ROC-AUCs of correlated models were compared with DeLong’s test [14, 15], with Holm adjustment over
190 the six comparisons. All other metrics were given 95% percentile bootstrap confidence intervals (2000 resamples of the hold-out patients).

3.7. Performance measures

We report accuracy, balanced accuracy, precision (positive predictive
195 value, PPV), recall (sensitivity), specificity, F1, the Matthews correlation coefficient (MCC) [35], ROC-AUC and average precision. Calibration was assessed with the Brier score, calibration-in-the-large (the intercept a of a logistic model with $\text{logit } \hat{p}$ as offset), the calibration slope b from $\text{logit } P(y = 1) = a' + b \text{ logit } \hat{p}$, and calibration curves [9]. A well-calibrated model has
200 $a = 0$ and $b = 1$. Calibration was evaluated both on the pooled out-of-fold predictions of the development set and on the hold-out set.

3.8. Threshold and decision-curve analysis

Instead of using the default 0.5 threshold, we pre-specified screening thresholds targeting 80%, 85%, 90% and 95% sensitivity. Each threshold
205 was chosen on the development-set out-of-fold predictions (averaged over the five repeats) and then applied unchanged to the hold-out set, where sensitivity, specificity, PPV, NPV and the proportion of patients flagged were estimated with bootstrap confidence intervals. Clinical utility was assessed by decision-curve analysis [16] on the hold-out set, using the net benefit

$$\text{NB}(p_t) = \frac{\text{TP}(p_t)}{n} - \frac{\text{FP}(p_t)}{n} \cdot \frac{p_t}{1 - p_t}, \quad (2)$$

210 over threshold probabilities p_t from 0.05 to 0.60, and compared with treating all or no patients.

3.9. Missing-data sensitivity analysis

We repeated the development-set comparison of logistic regression and RFECV + stacking (10-fold cross-validation repeated twice) with four imputation strategies, each fitted within the training folds: (i) median imputation;
215 (ii) median imputation plus a binary missingness indicator for every predictor with missing values, which lets the model use informative missingness [36]; (iii) iterative chained-equation imputation (scikit-learn `IterativeImputer`, Bayesian ridge, 20 iterations) [37]; and (iv) k -nearest-neighbour imputation
220 ($k = 10$) [38]. Differences from median imputation were tested with the corrected resampled t -test.

3.10. Explainability

We computed exact Shapley values [17] for the hold-out predictions of the final RFECV + stacking pipeline. The pipeline was treated as one function
225 of the eight raw clinical inputs, with 40 development-set patients as the background distribution. Explanations are therefore expressed in clinically measured variables, not in engineered or standardised features.

3.11. Implementation and reproducibility

The framework was implemented in Python 3.11 with scikit-learn 1.9 [39],
230 XGBoost 3.2, LightGBM 4.7 and SHAP. All random seeds were fixed (seed 42). Code, configuration, the fold-level results and the scripts that generate every table and figure are publicly available (see Data availability).

4. Results

4.1. Repeated cross-validation

235 Table 4 and Fig. 2 summarise the 5×10 -fold comparison on the development set. All learners except LightGBM and the single decision tree reached a mean ROC-AUC between 0.820 and 0.846, with overlapping corrected 95% confidence intervals. Stacking on all 16 features had the highest mean ROC-AUC (0.846, 95% CI 0.812–0.880), followed by stacking with the common
240 RFECV subset (0.843, 0.809–0.878) and logistic regression on all features (0.842, 0.809–0.875). The differences between the stacks and logistic regression were small and not statistically significant (+0.004 and +0.001; Holm-adjusted $p = 1.000$ for both). After Holm adjustment, only LightGBM (0.803 and 0.800 without and with RFECV; adjusted $p = 0.044$ and 0.020) and the
245 decision tree (0.770 and 0.772; $p = 0.025$ and 0.020) discriminated significantly worse than logistic regression. No configuration differed significantly from logistic regression in MCC.

At the default 0.5 threshold, the stacks had the highest mean sensitivity among the strong learners (0.763 without and 0.753 with RFECV, compared
250 with 0.744 for logistic regression and 0.655 for the SVM), at the cost of lower specificity (0.757 compared with 0.775 and 0.840). The Platt-calibrated SVM had the lowest mean Brier score (0.156; stacking + RFECV 0.165).

The common RFECV subset changed mean ROC-AUC by at most 0.003 for every learner while reducing the inputs from 16 to a mean of 12.0 features
255 (25% fewer). Logistic regression with its own coefficient-ranked RFECV kept 10.1 features on average and reached a ROC-AUC of 0.840 (0.808–0.873). It performed no better than logistic regression with the common, random-forest-ranked subset (0.842) or with all features.

Table 4: Development-set performance from 5×10-fold repeated cross-validation (50 folds; $n = 614$). Values are means; Nadeau–Bengio corrected 95% confidence intervals are shown in grey beneath. Δ AUC is the mean paired difference from logistic regression on all features, with Holm-adjusted corrected resampled t -test p -values in brackets. “+ RFECV” denotes the common, random-forest-ranked RFECV subset. Sensitivity and specificity use a 0.5 threshold.

Model	Features	ROC-AUC	MCC	Sens.	Spec.	Brier	Δ AUC (adj. p)
Stacking	16.0	0.846 (0.812–0.880)	0.505 (0.431–0.580)	0.763	0.757	0.164	+0.004 (1.000)
Stacking + RFECV	12.0	0.843 (0.809–0.878)	0.497 (0.418–0.575)	0.753	0.757	0.165	+0.001 (1.000)
Logistic regression	16.0	0.842 (0.809–0.875)	0.507 (0.438–0.576)	0.744	0.775	0.163	reference
Logistic regression + RFECV	12.0	0.842 (0.811–0.873)	0.492 (0.418–0.566)	0.717	0.782	0.163	0.000 (1.000)
LR (LR-ranked RFECV)	10.1	0.840 (0.808–0.873)	0.496 (0.426–0.567)	0.731	0.776	0.164	-0.002 (1.000)
SVM	16.0	0.840 (0.802–0.878)	0.505 (0.418–0.591)	0.655	0.840	0.156	-0.002 (1.000)
SVM + RFECV	12.0	0.837 (0.800–0.873)	0.483 (0.393–0.572)	0.643	0.830	0.158	-0.006 (1.000)
Extra trees + RFECV	12.0	0.835 (0.798–0.872)	0.474 (0.393–0.556)	0.693	0.786	0.161	-0.007 (1.000)
Extra trees	16.0	0.835 (0.795–0.874)	0.469 (0.377–0.560)	0.676	0.794	0.160	-0.008 (1.000)
Naive Bayes + RFECV	12.0	0.832 (0.797–0.867)	0.473 (0.393–0.554)	0.624	0.836	0.205	-0.010 (1.000)
Random forest	16.0	0.831 (0.795–0.868)	0.484 (0.390–0.578)	0.712	0.780	0.164	-0.011 (1.000)
Random forest + RFECV	12.0	0.831 (0.797–0.866)	0.492 (0.403–0.581)	0.718	0.781	0.164	-0.011 (1.000)
Gradient boosting	16.0	0.831 (0.798–0.863)	0.436 (0.353–0.518)	0.595	0.828	0.164	-0.011 (1.000)
Naive Bayes	16.0	0.830 (0.793–0.867)	0.494 (0.420–0.568)	0.626	0.852	0.205	-0.012 (1.000)
Gradient boosting + RFECV	12.0	0.830 (0.798–0.862)	0.441 (0.354–0.528)	0.601	0.827	0.164	-0.012 (1.000)
KNN	16.0	0.824 (0.781–0.866)	0.466 (0.375–0.558)	0.611	0.841	0.162	-0.019 (1.000)
XGBoost	16.0	0.821 (0.784–0.857)	0.436 (0.346–0.527)	0.611	0.816	0.173	-0.021 (0.954)
KNN + RFECV	12.0	0.821 (0.777–0.865)	0.458 (0.361–0.554)	0.604	0.839	0.163	-0.022 (1.000)
XGBoost + RFECV	12.0	0.820 (0.785–0.854)	0.440 (0.357–0.524)	0.615	0.817	0.172	-0.023 (0.586)
LightGBM	16.0	0.803 (0.764–0.842)	0.414 (0.321–0.508)	0.622	0.789	0.205	-0.039 (0.044)
LightGBM + RFECV	12.0	0.800 (0.765–0.836)	0.420 (0.336–0.505)	0.627	0.790	0.202	-0.042 (0.020)
Decision tree + RFECV	12.0	0.772 11 (0.731–0.813)	0.429 (0.348–0.510)	0.767	0.679	0.207	-0.070 (0.020)
Decision tree	16.0	0.770 (0.729–0.811)	0.431 (0.352–0.511)	0.774	0.675	0.210	-0.072 (0.025)

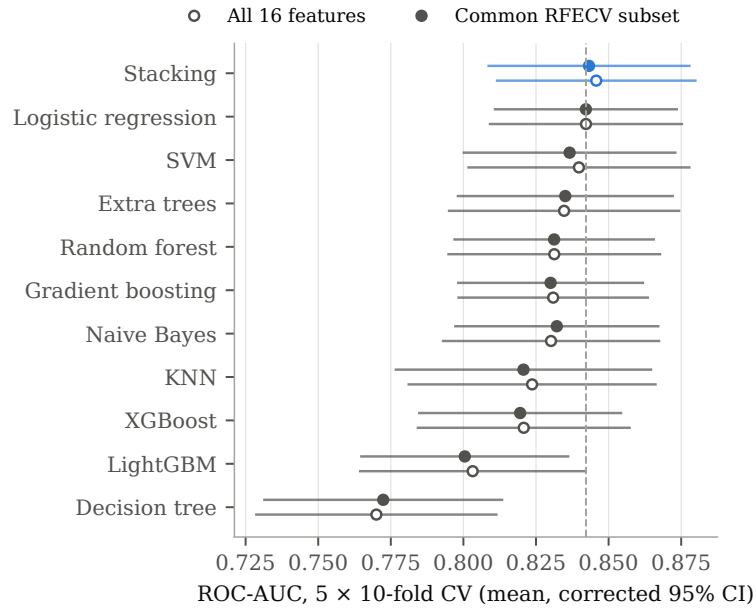


Figure 2: Cross-validated ROC-AUC of each learner with all 16 features (open circles) and with the common RFECV subset (filled circles). Bars are Nadeau–Bengio corrected 95% confidence intervals over 50 folds. The dashed line marks logistic regression on all features. The stacking ensemble is highlighted.

4.2. Selected features and their stability

260 Across the 50 outer folds, random-forest-ranked RFECV kept a median of 14 features (interquartile range 8–15; range 7–16). Seven features were kept in every fold: glucose, BMI and the engineered **Glucose_BMI**, **Glucose_Age**, **BMI_Age**, **HOMA_IR_surrogate** and **Pedigree_Age** (Fig. 3). The diabetes pedigree function was kept in 94% of folds. The thresholded flags were the
265 weakest candidates: **Obese** was kept in 6% of folds and **Hyperglycemic** in 38%. They largely duplicate information already carried by BMI and glucose.

Coefficient-ranked RFECV for logistic regression preferred a different, smaller subset (median 10.5 features; range 3–14). It kept glucose in every fold, and age, pregnancies and the pedigree function in more than 90% of
270 folds. It rarely kept blood pressure (2%) and never kept skin-fold thickness. The core glycaemic, adiposity and family-history signal was therefore stable across both selectors, but the exact subset depended on the ranking criterion.

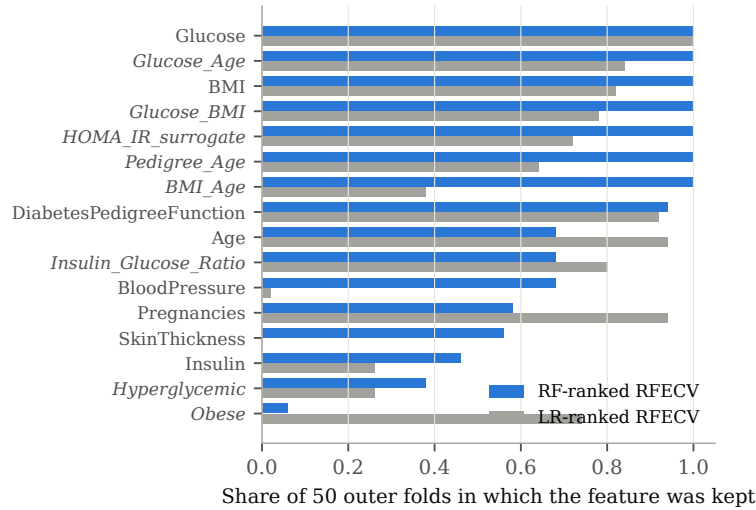


Figure 3: Proportion of the 50 outer cross-validation folds in which each candidate feature was retained by random-forest-ranked RFECV (the common protocol) and by coefficient-ranked RFECV for logistic regression. Engineered features are shown in italics.

4.3. Hold-out performance

275 On the 154 hold-out patients (Table 5, Fig. 4), the RFECV + stacking pipeline reached a ROC-AUC of 0.822 (95% CI 0.756–0.888), sensitivity 0.815

(0.709–0.915), specificity 0.740 (0.649–0.823), MCC 0.532 (0.403–0.658) and F1 0.710. It correctly identified 44 of 54 patients with diabetes. Among the evaluated configurations it had the highest observed sensitivity, F1 and MCC on this split. However, the confidence intervals of all models overlapped widely. No ROC-AUC difference between the RFECV + stacking pipeline and any comparator was statistically significant (DeLong test; differences from -0.004 to $+0.013$; smallest Holm-adjusted $p = 0.89$). Random forest + RFECV (0.826) and XGBoost + RFECV (0.824) had slightly higher hold-out ROC-AUC, and average precision ranged from 0.683 (logistic regression) to 0.711 (random forest + RFECV).

Table 5: Hold-out performance ($n = 154$, 54 with diabetes) at a 0.5 threshold. 95% confidence intervals are shown in grey beneath: DeLong for ROC-AUC, percentile bootstrap (2000 resamples) otherwise. AP, average precision. The last column gives the ROC-AUC difference of RFECV + stacking minus the comparator, with Holm-adjusted DeLong p -values.

Model	ROC-AUC	Sensitivity	Specificity	MCC	F1	AP	Δ AUC (adj. p)
Stacking + RFECV	0.822 (0.756–0.888)	0.815 (0.709–0.915)	0.740 (0.649–0.823)	0.532 (0.403–0.658)	0.710 (0.615–0.797)	0.696	–
Stacking	0.819 (0.752–0.885)	0.796 (0.688–0.904)	0.720 (0.631–0.804)	0.494 (0.362–0.625)	0.688 (0.592–0.778)	0.702	+0.003 (1.000)
Logistic regression	0.811 (0.744–0.878)	0.741 (0.625–0.862)	0.740 (0.649–0.825)	0.464 (0.325–0.606)	0.667 (0.564–0.763)	0.683	+0.011 (1.000)
LR + LR-ranked RFECV	0.809 (0.741–0.876)	0.722 (0.608–0.846)	0.740 (0.649–0.825)	0.447 (0.309–0.591)	0.655 (0.551–0.753)	0.684	+0.013 (0.892)
Random forest + RFECV	0.826 (0.761–0.892)	0.685 (0.564–0.804)	0.760 (0.673–0.840)	0.434 (0.290–0.573)	0.643 (0.537–0.739)	0.711	-0.004 (1.000)
XGBoost + RFECV	0.824 (0.758–0.891)	0.648 (0.523–0.776)	0.820 (0.743–0.888)	0.470 (0.321–0.605)	0.654 (0.543–0.752)	0.697	-0.003 (1.000)
SVM	0.810 (0.740–0.880)	0.537 (0.407–0.672)	0.830 (0.755–0.896)	0.383 (0.226–0.534)	0.580 (0.457–0.692)	0.684	+0.012 (1.000)

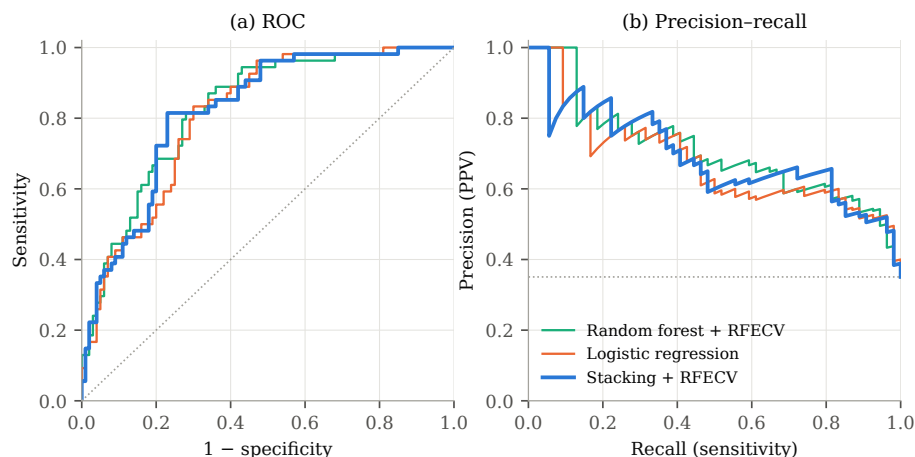


Figure 4: Hold-out ROC (a) and precision-recall (b) curves for RFECV + stacking, logistic regression and random forest + RFECV. The dotted lines show chance performance and the outcome prevalence (0.35).

4.4. Calibration

All class-weighted models overestimated risk (Table 6, Fig. 5). On the pooled development-set out-of-fold predictions, the calibration intercept was -0.616 for RFECV + stacking and -0.623 for logistic regression. This is almost exactly the logit of the outcome prevalence, $\text{logit}(0.349) = -0.62$. That is the shift expected when balanced class weights train a model as if the classes were equally frequent. The calibration slope of the stack was close to one (1.002), so its spread of risks was appropriate and only its level was shifted. The Platt-calibrated SVM, whose probabilities are refitted without class weights, was well calibrated (intercept -0.029 , slope 0.985) and had the lowest Brier score. On the hold-out set, the stack's intercept (-0.624) was unchanged and its slope fell to 0.866, compared with 0.756 for logistic regression.

Table 6: Calibration. Development values use pooled out-of-fold predictions (614×5); hold-out Brier scores have bootstrap 95% CIs. Ideal values: intercept 0, slope 1.

Model	Development (out-of-fold)			Hold-out		
	Brier	Intercept	Slope	Brier (95% CI)	Intercept	Slope
Stacking + RFECV	0.165	-0.616	1.002	0.177 (0.143–0.213)	-0.624	0.866
Logistic regression	0.163	-0.623	0.899	0.181 (0.147–0.219)	-0.629	0.756
Random forest + RFECV	0.164	-0.406	0.823	0.166 (0.134–0.201)	-0.338	0.776
SVM	0.156	-0.029	0.985	0.169 (0.135–0.205)	+0.004	0.844

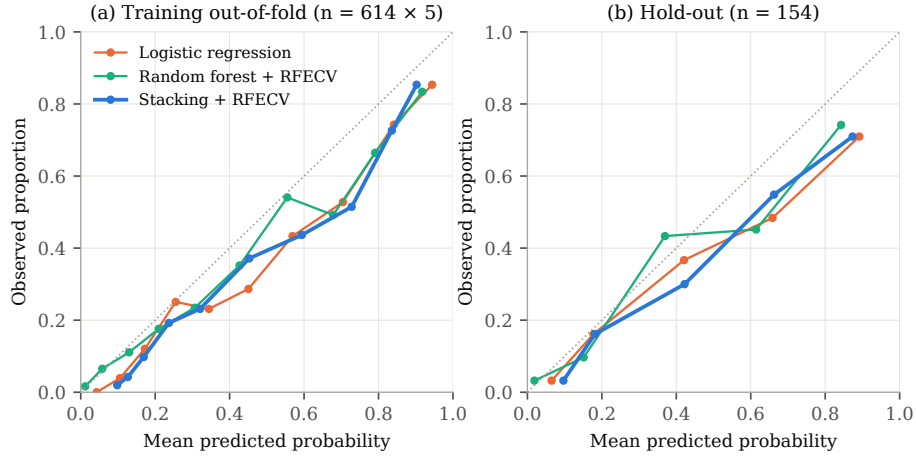


Figure 5: Calibration curves (deciles of predicted risk) on (a) pooled development-set out-of-fold predictions and (b) the hold-out set (quintiles). Points below the diagonal indicate overestimated risk.

4.5. Screening thresholds and clinical utility

300 Thresholds fixed on development-set out-of-fold predictions transferred reasonably to the hold-out set (Table 7). For RFECV + stacking, the 80%, 85% and 90% sensitivity targets gave hold-out sensitivities of 0.815, 0.852 and 0.907 and specificities of 0.66, 0.63 and 0.56, flagging 51%–60% of patients for confirmatory testing. The 95% target gave a hold-out sensitivity of 0.926
305 (0.848–0.983), below target. With only 54 positive patients, each missed case changes sensitivity by 1.9 percentage points. Logistic regression gave very similar operating points; at the 95% target it reached 0.963 with specificity 0.53. NPV was 0.87–0.96 across all thresholds and both models.

In decision-curve analysis (Fig. 6), all three models had higher net benefit
310 than treating all or no patients across threshold probabilities from 0.10 to 0.55; logistic regression fell to zero net benefit at 0.60. The three curves were nearly identical up to a threshold of 0.45 (differences in net benefit of at most 0.03). Above 0.50, the stack retained slightly more net benefit than logistic regression, but this region is based on few patients.

Table 7: Sensitivity-targeted operating points. Thresholds were chosen on development-set out-of-fold predictions and applied unchanged to the hold-out set. Bootstrap 95% CIs are shown in grey beneath. “Flagged” is the proportion of hold-out patients referred for confirmatory testing.

Model	Target	Threshold	Sensitivity	Specificity	PPV	NPV	Flagged
Stacking + RFECV	80%	0.433	0.815 (0.709–0.915)	0.660 (0.559–0.747)	0.564 (0.453–0.670)	0.868 (0.790–0.940)	50.6%
Stacking + RFECV	85%	0.348	0.852 (0.754–0.942)	0.630 (0.529–0.723)	0.554 (0.442–0.658)	0.887 (0.809–0.957)	53.9%
Stacking + RFECV	90%	0.276	0.907 (0.821–0.981)	0.560 (0.459–0.653)	0.527 (0.420–0.622)	0.918 (0.839–0.983)	60.4%
Stacking + RFECV	95%	0.212	0.926 (0.848–0.983)	0.520 (0.423–0.621)	0.510 (0.408–0.606)	0.929 (0.855–0.984)	63.6%
Logistic regression	80%	0.413	0.852 (0.754–0.942)	0.650 (0.551–0.738)	0.568 (0.458–0.671)	0.890 (0.814–0.958)	52.6%
Logistic regression	85%	0.345	0.852 (0.754–0.942)	0.620 (0.520–0.710)	0.548 (0.438–0.654)	0.886 (0.806–0.956)	54.5%
Logistic regression	90%	0.284	0.889 (0.804–0.968)	0.580 (0.480–0.674)	0.533 (0.427–0.633)	0.906 (0.829–0.970)	58.4%
Logistic regression	95%	0.212	0.963 (0.903–1.000)	0.530 (0.431–0.630)	0.525 (0.419–0.624)	0.964 (0.903–1.000)	64.3%

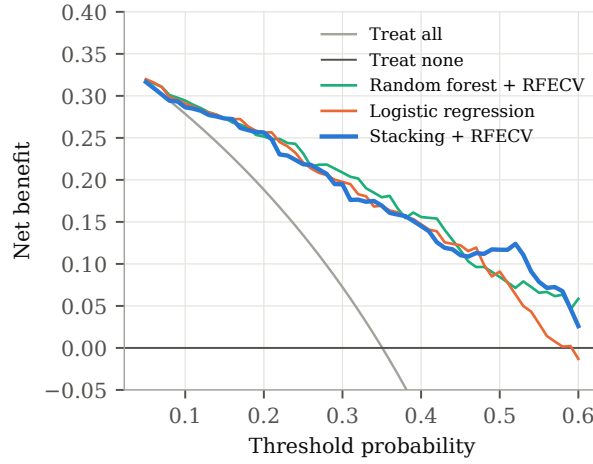


Figure 6: Decision curves on the hold-out set. Net benefit is shown against the threshold probability at which a patient would be referred for confirmatory testing.

315 4.6. Sensitivity to missing-data handling

The conclusions were robust to the imputation strategy (Table 8). With iterative or k -NN imputation, ROC-AUC changed by at most +0.005 relative to median imputation for both logistic regression and RFECV + stacking (all $p \geq 0.45$). Adding missingness indicators did not help. It lowered the ROC-AUC of logistic regression slightly (-0.010 , $p = 0.39$) and left RFECV + stacking unchanged (-0.002 , $p = 0.50$), and RFECV retained on average only 1.2 of the five indicators. Under every strategy, stacking and logistic regression remained within 0.005 of each other.

Table 8: Missing-data sensitivity analysis on the development set (10-fold cross-validation repeated twice; 20 folds). ROC-AUC values are means with corrected 95% CIs. Δ AUC is the paired difference from median imputation for the same model, with the corrected resampled t -test p -value.

Model	Imputation	Features	ROC-AUC (95% CI)	MCC	Sens.	Brier	Δ AUC (p)
Logistic regression	Median	16.0	0.846 (0.813–0.878)	0.514	0.743	0.162	reference
Logistic regression	Median + indicators	21.0	0.836 (0.788–0.884)	0.484	0.727	0.166	-0.010 (0.387)
Logistic regression	Iterative (MICE-style)	16.0	0.847 (0.814–0.879)	0.517	0.753	0.162	+0.001 (0.712)
Logistic regression	KNN ($k = 10$)	16.0	0.847 (0.813–0.880)	0.505	0.743	0.162	+0.001 (0.607)
Stacking + RFECV	Median	11.3	0.844 (0.804–0.883)	0.498	0.752	0.164	reference
Stacking + RFECV	Median + indicators	13.3	0.841 (0.799–0.884)	0.496	0.748	0.166	-0.002 (0.503)
Stacking + RFECV	Iterative (MICE-style)	14.0	0.848 (0.815–0.880)	0.512	0.762	0.163	+0.004 (0.594)
Stacking + RFECV	KNN ($k = 10$)	13.6	0.849 (0.814–0.884)	0.501	0.759	0.163	+0.005 (0.448)

4.7. Explanation of predictions

Glucose dominated the SHAP explanations of the RFECV + stacking pipeline (mean $|\text{SHAP}| = 0.170$ on the probability scale), followed by BMI (0.074), age (0.055), the diabetes pedigree function (0.038) and pregnancies (0.028) (Fig. 7). Insulin, skin-fold thickness and blood pressure contributed little (at most 0.015). Higher glucose, BMI, age and pedigree values consistently increased predicted risk, which agrees with established clinical risk factors.

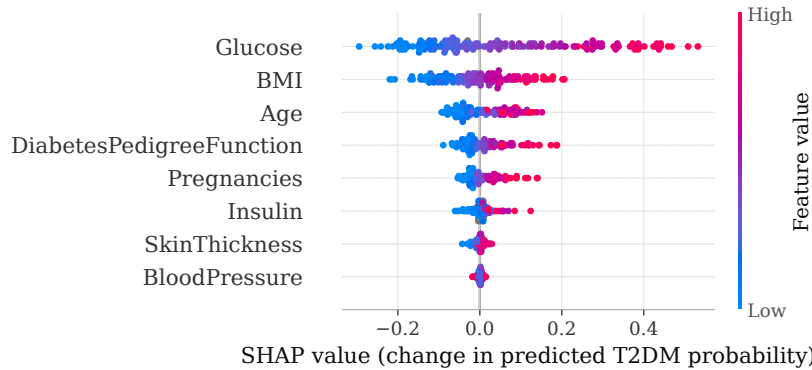


Figure 7: SHAP values of the hold-out predictions of the RFECV + stacking pipeline, expressed in the eight raw clinical inputs. Each point is one patient; colour shows the feature value.

5. Discussion

5.1. Principal findings

Under a validation design in which every data-dependent step was re-fitted inside each split, combining RFECV with an eight-learner stacking ensemble did not improve discrimination over regularised logistic regression on the PIMA cohort. Across 50 cross-validation folds, the stacks and logistic regression differed by less than 0.005 in ROC-AUC, and no difference approached significance after accounting for correlated folds and multiple comparisons. The same held on the hold-out set, where every DeLong comparison was non-significant. The higher hold-out sensitivity, F1 and MCC of RFECV + stacking (0.815, 0.710 and 0.532) are best read as the highest observed values on one 154-patient split, not as evidence of superiority. Its bootstrap confidence intervals overlap those of every comparator.

This null result is itself informative. It agrees with the systematic finding that flexible ML methods offer no average discrimination benefit over logistic regression in clinical prediction [10]. It also contrasts with the very high accuracies (0.90–0.98) sometimes reported for complex pipelines on the same 768 patients [23, 24]. We cannot reproduce or audit those studies. However, a dataset with eight routine variables, a 35% prevalence and a cross-validated ROC-AUC ceiling near 0.85 leaves little room for such gains. Validation design, especially preprocessing and feature selection outside the resampling loop [6, 8], should be the first explanation considered for them.

5.2. What RFECV and stacking contribute

RFECV reduced the inputs by about a quarter with no loss of discrimination and identified a stable core of glycaemic, adiposity and family-history information, consistent with the SHAP analysis. The engineered interactions (Glucose_BMI, Glucose_Age, BMI_Age, HOMA_IR_surrogate, Pedigree_Age) were consistently preferred over thresholded flags and over raw blood pressure or skin-fold thickness. Two caveats apply. First, the selected size varied widely between folds (7–16 features) because the RFECV score curve is flat above about six features. A one-standard-error rule or stability selection [40] would yield smaller and more reproducible subsets. Second, the subset depended on the ranking estimator: coefficient-ranked RFECV for logistic regression chose different features than random-forest-ranked RFECV. The common protocol used here ensures a fair, identical selection procedure

across learners, but it is not a learner-specific optimum, and results for individual learners should be read with that in mind.

Stacking added robustness rather than discrimination. It was never significantly worse than the best single learner, it did not require choosing a learner in advance, and at the default threshold it operated at higher sensitivity. Its meta-learner can also expose redundant base learners. Against these advantages stand higher computational cost (one RFECV + stacking fit took about 45 s, compared with milliseconds for logistic regression) and reduced transparency. On cohorts of this size, logistic regression remains a strong and simpler default.

5.3. Calibration and thresholds matter more than the choice of learner

The largest practical difference between configurations was not discrimination but calibration. Balanced class weights, which are widely used to raise sensitivity on imbalanced data, shifted predicted risks upwards by almost exactly the logit of the prevalence. The calibration slope stayed near one. The ranking of patients is therefore unaffected, but the predicted probabilities cannot be read as absolute risks without recalibration [9]. A simple intercept correction or Platt rescaling without class weights, fitted on out-of-fold predictions, would remove this offset. The well-calibrated SVM in our comparison shows the effect directly. Reporting only discrimination metrics, as most PIMA studies do, would have hidden this issue entirely.

Similarly, the default 0.5 threshold is arbitrary for screening. Thresholds chosen on development data for 80%–90% sensitivity transferred to the hold-out set within sampling error, at a cost of referring roughly half to 60% of patients for confirmatory testing. Decision curves showed positive net benefit over the clinically plausible threshold range. However, all three models gave nearly identical net benefit, so the operating point matters more for clinical use than the choice between them.

5.4. Missing data

Almost half of the insulin values and 30% of the skin-fold values were missing, so a reviewer might expect missingness itself to carry predictive information [36]. In this cohort it did not. Missingness indicators slightly reduced discrimination, and iterative or k -NN imputation gave the same results as the simpler median imputation. The low SHAP contributions of insulin and skin-fold thickness agree with this. Whatever information these partially observed variables carry is largely redundant with glucose and

BMI. Median imputation fitted within training folds is therefore an adequate choice here, although this may not hold in cohorts where testing is ordered
405 selectively.

5.5. Limitations

- **One small, specific cohort.** PIMA comprises 768 women of one ethnic group from a single study. The results may not transfer to men, other populations or contemporary diagnostic criteria based on HbA1c.
410 With 268 events and 16 candidate predictors, the cohort is also small for flexible model development [41].
- **No external validation.** Repeated cross-validation and a hold-out set estimate internal performance only. Validation on independent cohorts, such as population surveys with OGTT measurements, is needed before
415 any clinical use.
- **One hold-out split.** With 54 positive hold-out patients, confidence intervals are wide. We therefore base inference on the 50-fold repeated cross-validation and treat hold-out results as confirmatory.
- **Fixed hyperparameters.** Learners were not tuned, to avoid an additional selection loop. Nested tuning could improve individual learners slightly, although the cross-validated ceiling suggests limited headroom.
420
- **Post-load measurements.** PIMA glucose and insulin are 2-h OGTT values, so `HOMA_IR_surrogate` is not a clinical HOMA-IR, and the model applies to settings where an OGTT has already been performed.

5.6. Future work

Future work should (i) validate the pipeline externally on larger, more diverse cohorts; (ii) incorporate recalibration and threshold selection into the nested pipeline; (iii) compare RFECV with stability selection and with learner-specific nested RFECV; and (iv) assess fairness and transportability
430 across sex and ethnicity, as recommended by TRIPOD+AI [30].

6. Conclusion

We evaluated a leakage-controlled framework that combines RFECV with a heterogeneous stacking ensemble for T2DM prediction, using repeated

cross-validation with corrected inference, an untouched hold-out set, calibration, threshold and decision-curve analysis, and SHAP explanations. RFECV
435 produced compact feature sets built on glucose, BMI, family history and their interactions, without loss of discrimination. The stacking ensemble achieved the highest observed hold-out sensitivity (0.815), F1 (0.710) and MCC (0.532) on this split. However, across 50 cross-validation folds and
440 on the hold-out set, it did not discriminate significantly better than regularised logistic regression (cross-validated ROC-AUC 0.843 versus 0.842). Class weighting produced a systematic overestimation of risk that discrimination metrics do not reveal. On small clinical tabular datasets, increased ensemble complexity therefore does not by itself improve prediction. The
445 main value of such a framework lies in rigorous, leakage-free evaluation, sensitivity-oriented operating points and transparent reporting of calibration and clinical utility.

CRedit authorship contribution statement

Subrahmanyam Tanala: Conceptualization, Methodology, Software,
450 Formal analysis, Investigation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this
455 paper.

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Ethics statement

460 This study used only the publicly available, de-identified PIMA Indians Diabetes dataset. No new data were collected from human participants, so ethical approval and informed consent were not required.

Data and code availability

The PIMA Indians Diabetes dataset is publicly available. All code, configuration files, fold-level results and the scripts that regenerate every table and figure in this article are available at <https://github.com/subrahmanyanmtanala-rgb/T2D>. The full analysis can be reproduced with `python -m t2d.analysis -stage cv, -stage holdout and -stage missing`, followed by `python paper/make_assets.py`.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Claude (Anthropic) to assist with software development, statistical analysis code and drafting of the manuscript text. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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